PARISHRAM 2.0

CLASS 12TH 2026

Relations and Functions

MATHS **LECTURE-6**

BY-MANDAR SIR



TOPICS TO BE COVERED

1 More Questions on Types of Functions

Check one-one
onto

Greatest Integer

$$f(x) = [x]$$

input	output
$(0 - 0.9)$	$= 0$

$$f(x) = [x]$$

$$= [0.8] = 0$$

$$= [0.4] = 0$$

Comment

$[11.7] = 11$

$f(x) = [x]$

$[2.3] = 2$

$[1.9] = 1$

Question

Prove that the Greatest Integer Function $f : R \to R$, given $f(x) = [x]$, is neither one-one nor onto, where $[x]$ denotes the greatest integers less than or equal to x.

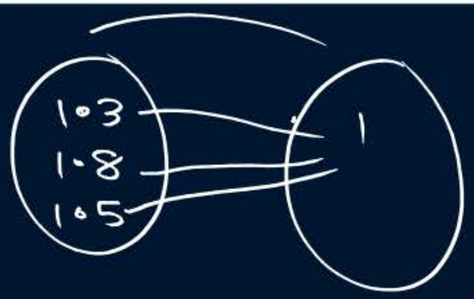
For into

$f(x) = [x]$

If $x_1 = 1.3 \longrightarrow f(x) = [1.3] = 1$

$x_2 = 1.8 \qquad f(x) = [1.8] = 1$

One-one f^n ✗



For onto

$f(x) = [x]$

$y = [x]$

$y = 1.7$ ✗

Range ↳ integers

Co-domain → Real values

Range ≠ Co-domain

Modulus fn

$f(x) = |x|$

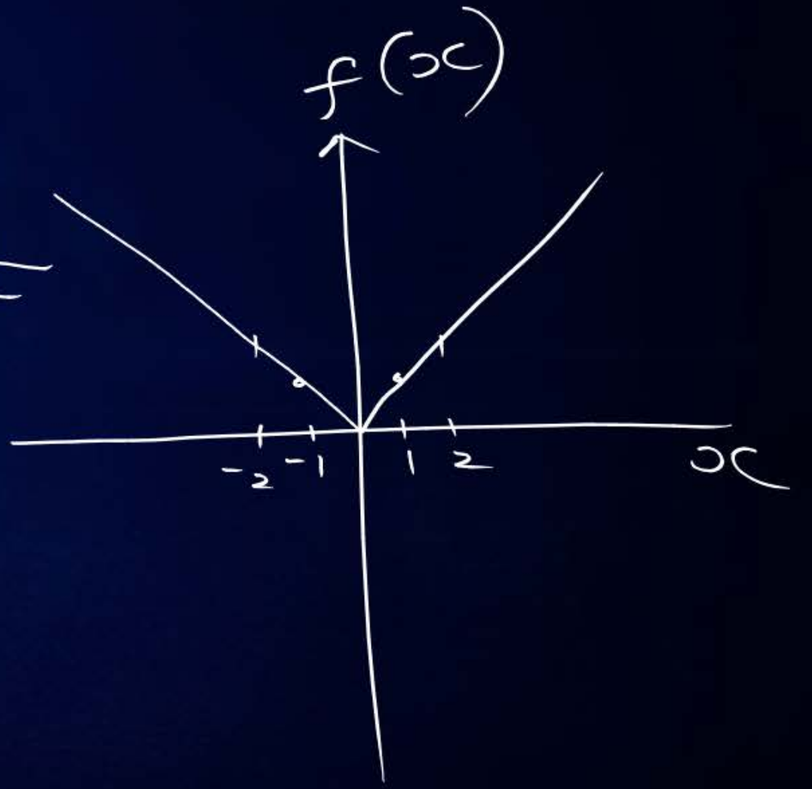
Input → −, +

Output → +

$|x|$

$|-3| = +3$

$|+3| = +3$



Question

Show that the Modulus Function $f : R \to R$, given by $f(x) = |x|$, is neither one-one nor onto, where $|x|$ is x, if x if positive or 0 and $|x|$ is $-x$, if x is negative.

$f(x) = |x|$

$|-3| \to +3$

$|+3| \to +3$

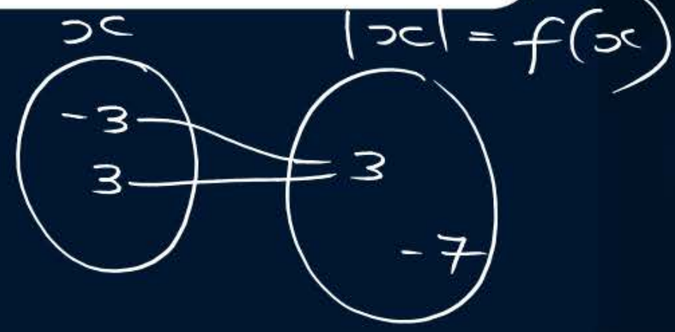
One-one function ✗

$y = |x|$

Assume $y = -7$

But output can't be negative

} Not a onto function



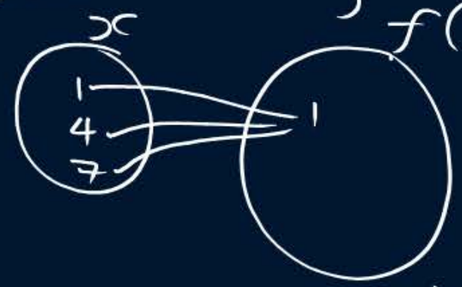
Question

Show that the Signum Function $f: \mathrm{R} \to \mathrm{R}$, given by

$$f(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

is neither one-one nor onto.

One-One f^n



One-One f^n ✗

$y = f(x)$

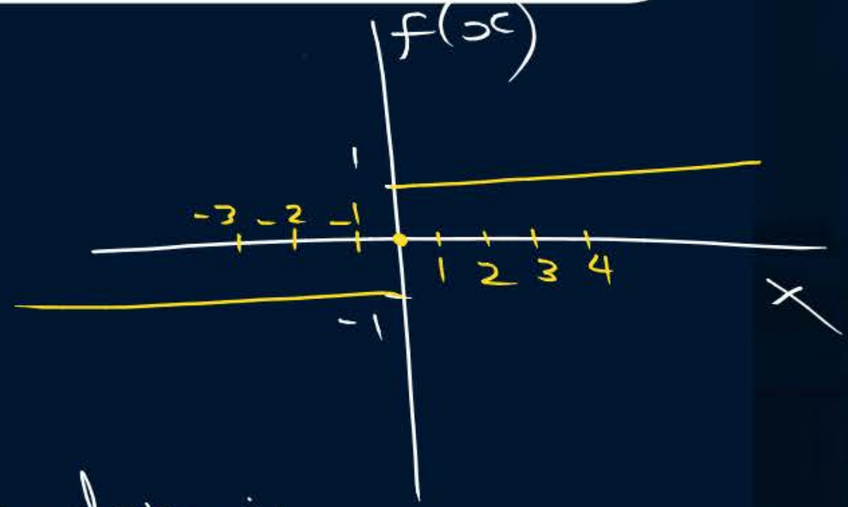
Assume

$y = \sqrt{11}$

Output ??

Range ≠ Co-domain

$\{1, 0, -1\}$ Onto ✗



Question

In each of the following cases, state whether the function is one-one, onto or bijective. Justify your answer.

(i) $f : \mathbf{R} \to \mathbf{R}$ **defined by** $f(x) = 3 - 4x$

$f(x) = 3 - 4x$

One-one

$f(x_1) = f(x_2)$

$3 - 4x_1 = 3 - 4x_2$

$-4x_1 = -4x_2$

$x_1 = x_2$

One-one ✓

Onto

$f(x) = 3 - 4x$

$y = 3 - 4x$

$4x = 3 - y$

$x = \frac{3 - y}{4}$

$y \to$ Real Value put

$x \to$ Real ✓

Range = Co-domain

Question

Let A = R – {3} and B = R – {1}. Consider the function $f : A \to B$ defined by $f(x) = \frac{(x-2)}{(x-3)}$ is f one-one and onto? Justify your

State whether the function f is bijective. Justify your answer.

$f(x) = \frac{x-2}{x-3}$

One-One

$f(x_1) = f(x_2)$

$\frac{x_1-2}{x_1-3} = \frac{x_2-2}{x_2-3}$

$(x_1-2)(x_2-3) = (x_2-2)(x_1-3)$

$x_1x_2 - 3x_1 - 2x_2 + 6 = x_1x_2 - 3x_2 - 2x_1 + 6$

$-3x_1 - 2x_2 = -3x_2 - 2x_1$

$-3x_1 + 2x_1 = -3x_2 + 2x_2$

$-x_1 = -x_2$

$x_1 = x_2$ ⟹ One One f^n ✓

Onto f^n

$$f(x) = \frac{x-2}{x-3}$$

$$y = \frac{x-2}{x-3}$$

$$y(x-3) = x-2$$

$$xy - 3y = x - 2$$

$$xy - x = 3y - 2$$

$$x(y-1) = 3y - 2$$

$$x = \frac{3y-2}{y-1}$$

$y \rightarrow$ Real Value (Not 1)

x (input) $\rightarrow$ Real Value (Not 3)

Co-domain = Range

Onto function

Question

Let $f : \mathbf{N} \to \mathbf{N}$ be defined by $f(n) = \begin{cases} \frac{n+1}{2}, & \text{if } n \text{ is odd} \\ \frac{n}{2}, & \text{if } n \text{ is even} \end{cases}$ for all $n \in \mathbf{N}$.

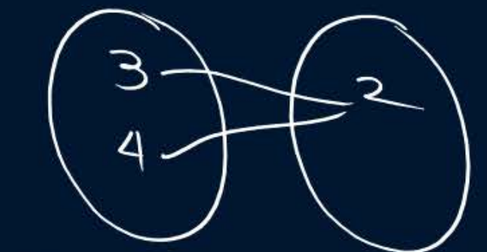
State whether the function f is bijective. Justify your answer.

If $n = 3$ (odd)

$\frac{n+1}{2} = \frac{3+1}{2} = \frac{4}{2} = 2$

If $n = 4$

$\frac{n}{2} = \frac{4}{2} = 2$



One-One ✗

Onto (case 1)

$y = \frac{n+1}{2}$ (odd)

$2y = n + 1 \rightarrow 2y - 1 = n$

$y \rightarrow$ Natural No

$n \rightarrow$ odd

Case 2

$y = \frac{n}{2}$

THANK YOU